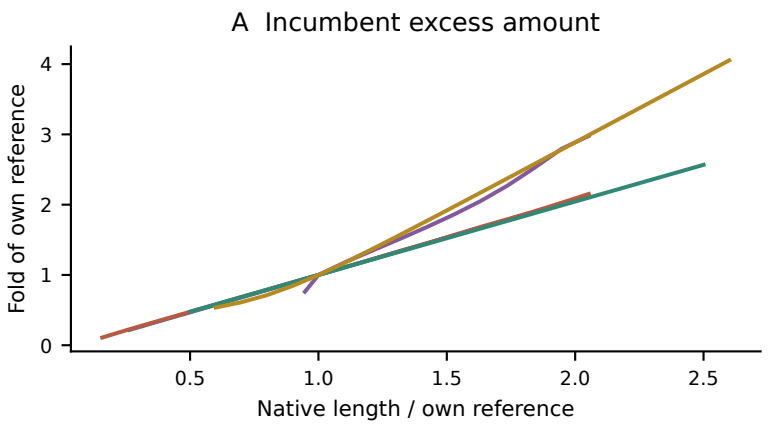
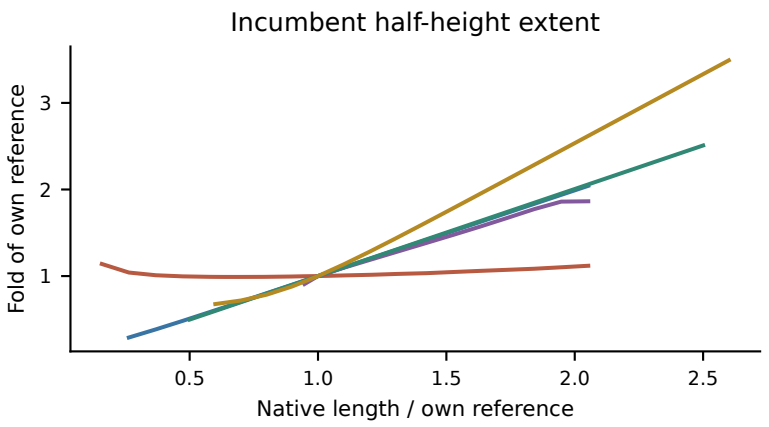
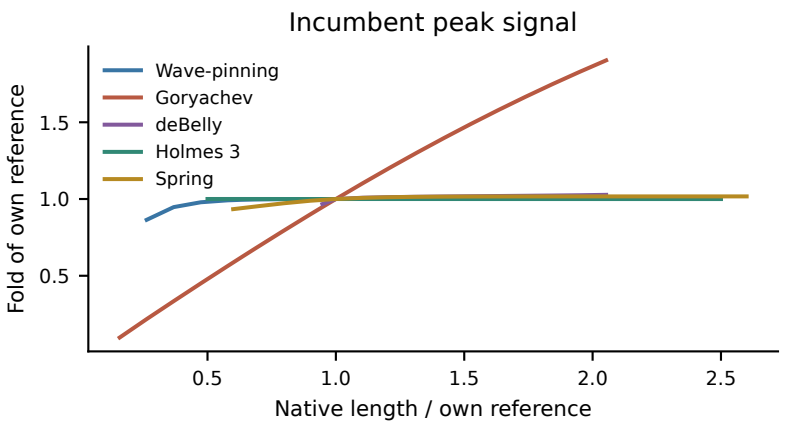
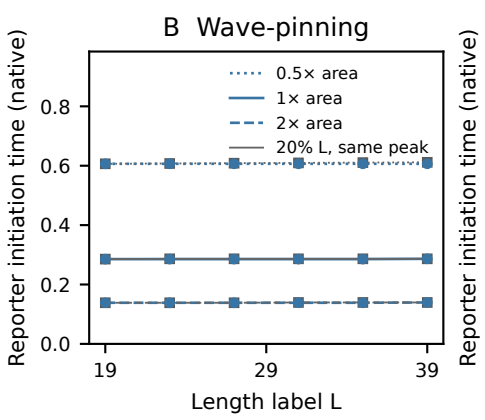
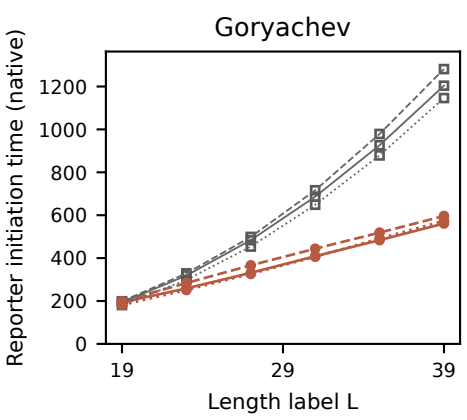
Figure 2. Shape and timing distinguish replacement responses
AUDITED FINAL

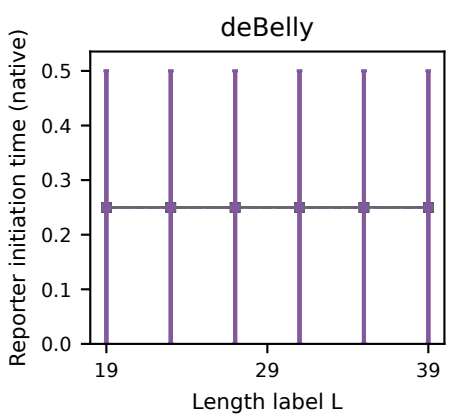


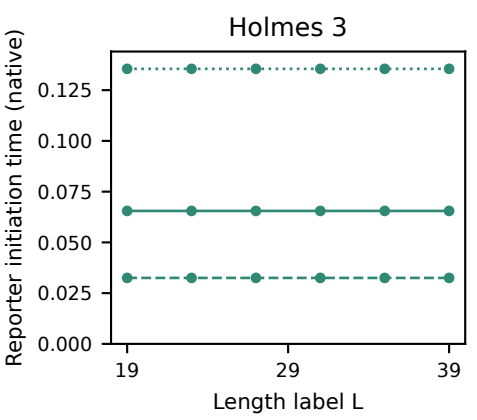


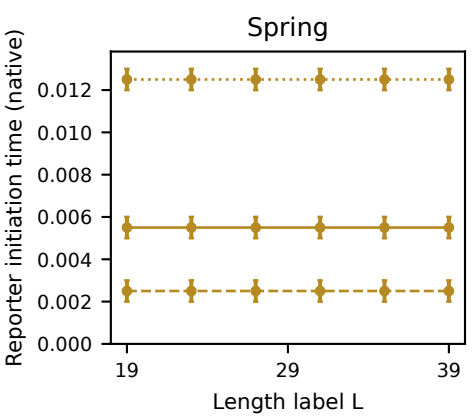


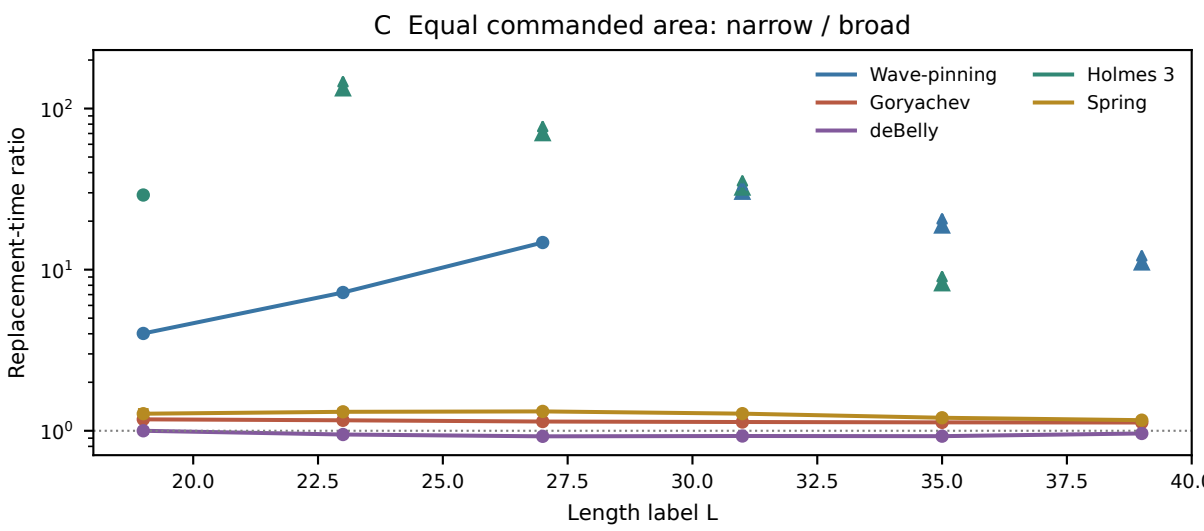


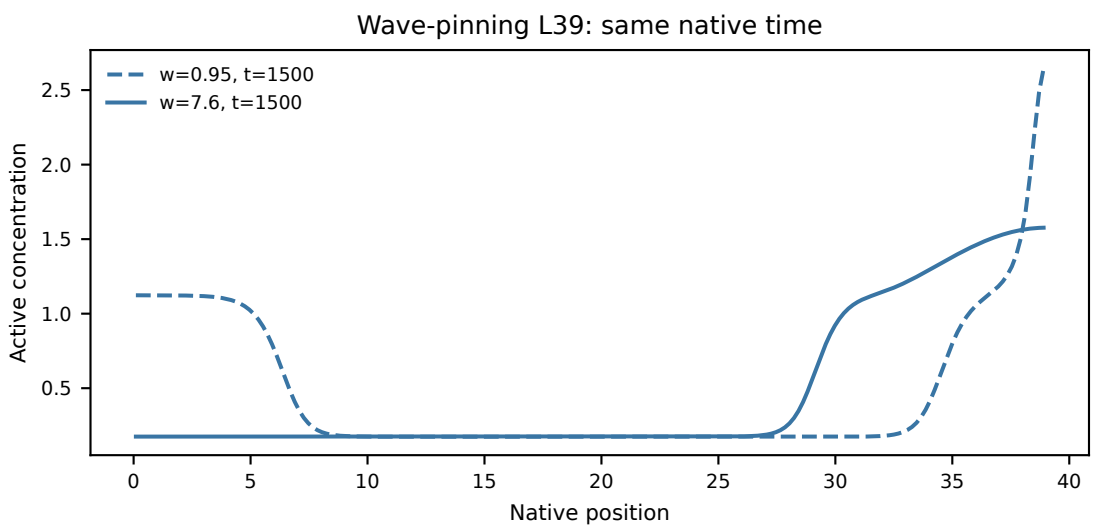


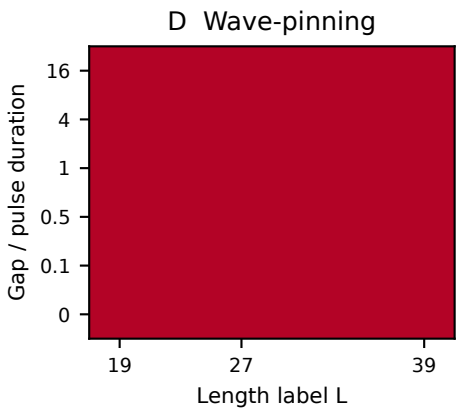


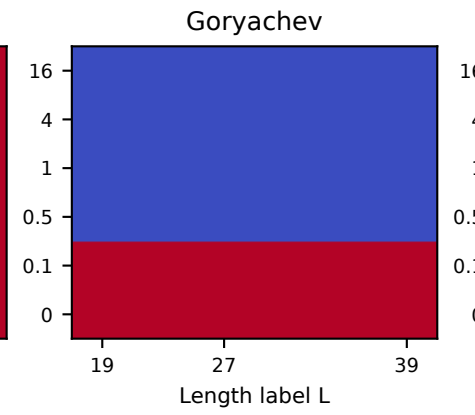


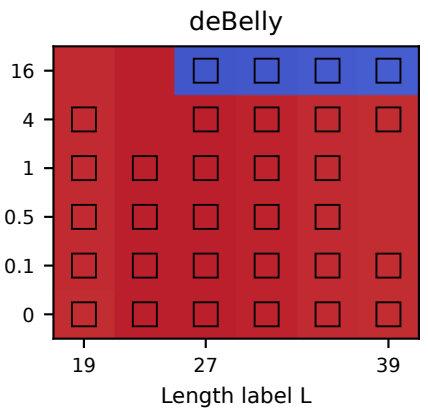


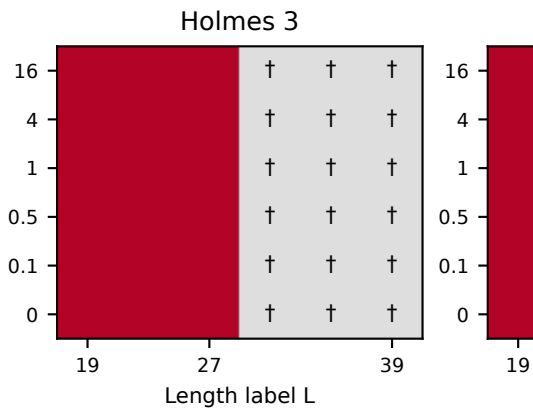


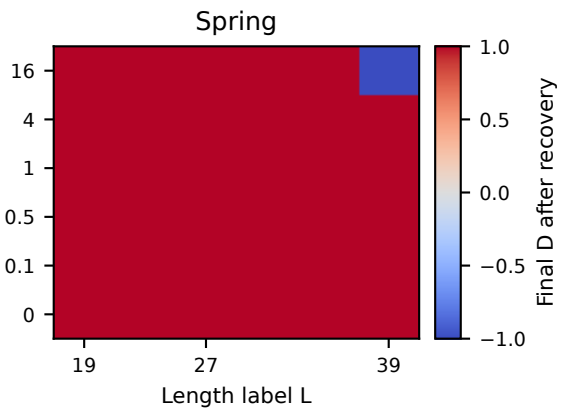


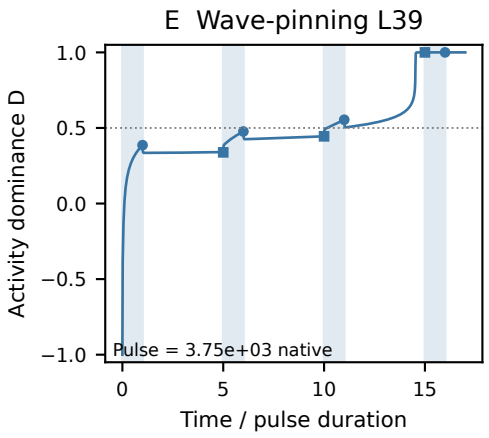


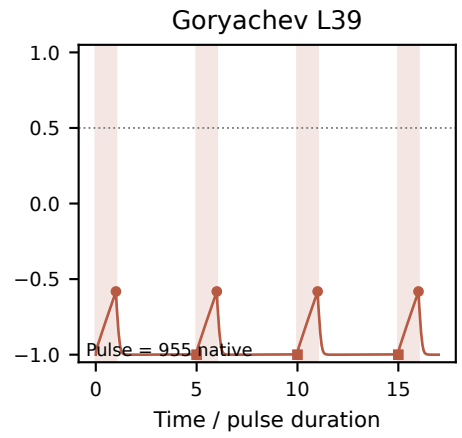


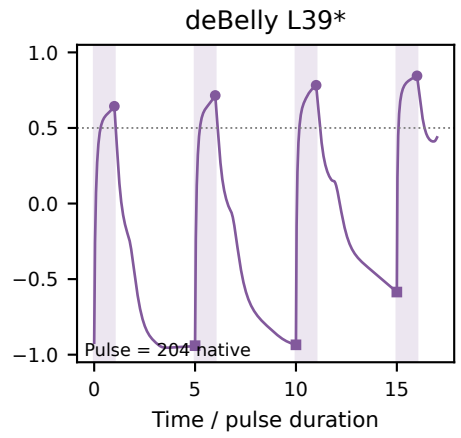


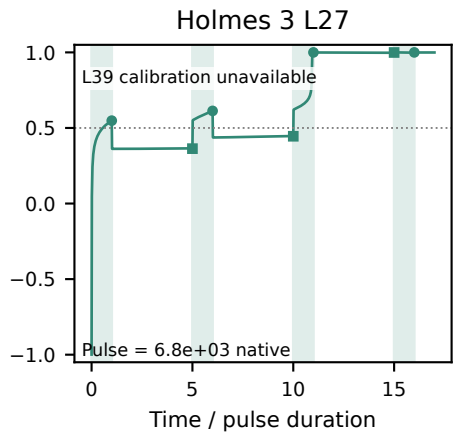


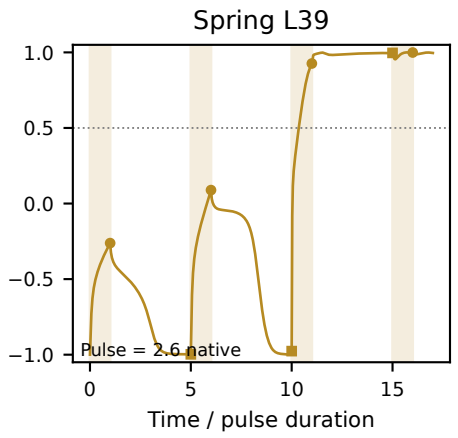












F Signatures tied to measurements; causal interpretation requires native-field support

Model	Trace L	Initiation L39/L19	Narrow / broad, L19 → L39	Pulse 1 → gap 1: D	Final D
Wave-pinning	L39	1–1.01	3.99–4.05 → >11.1	0.39 → 0.34	1.00
Goryachev	L39	2.87–2.92	1.17–1.18 → 1.12–1.13	-0.58 → -1.00	-1.00
deBelly*	L39	both in (0, 0.5]	0.962–1.04 → 0.955–0.97	0.64 → -0.94	0.92
Holmes 3	L27	0.985–1.02	29–29.1 → uninformative	0.55 → 0.36	1.00
Spring	L39	0.833–1.2	1.2–1.36 → 1.15–1.18	-0.26 → -1.00	1.00

Shape: beam width at fixed input area • Timing: equal total ON pulse trains • Reporter: activity share after each gap

↑ triangles: finite-cap ratio lower bounds; bounds cannot be ranked. †: no calibration; ×: low activity; □: failed field tail.

* deBelly later-gap timing and endpoint fields have numerical qualifications (S5). Native clocks differ; optical dose is uncalibrated.